YOUR TECHNOLOGY & INNOVATION PARTNER

Advanced AI Training - Session 2

Explainable AI (XAI)

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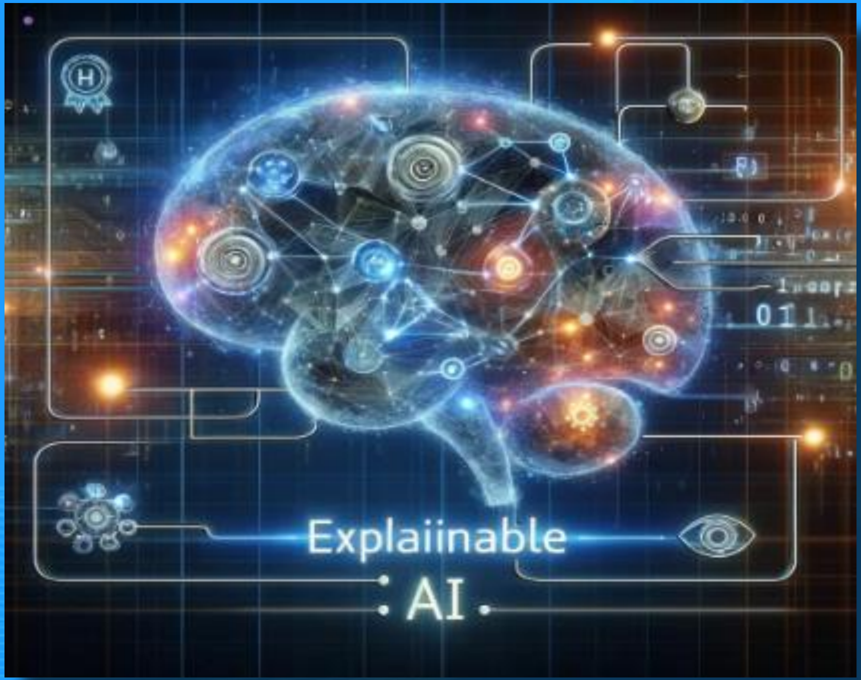
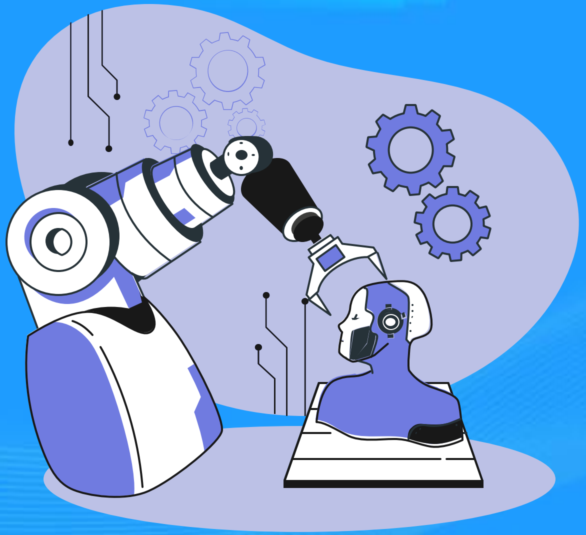
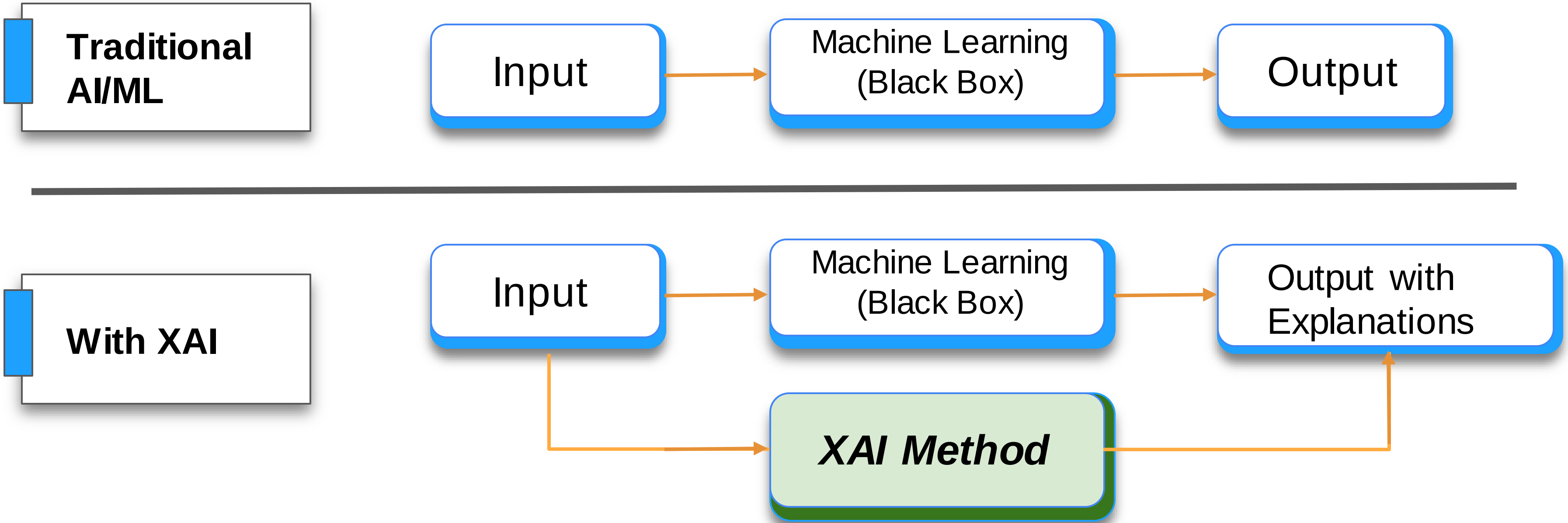


Introduction of XAI



What is Explainable AI (XAI)?

Explainable artificial intelligence (XAI) is a set of processes and methods that allows human users to comprehend and trust the results and output created by machine learning algorithms.



Source: IBM (Link)

Explainable AI (XAI) - Wordcloud

Transparency

Interpretability

Ethical & Responsible AI

Key Related Concepts of XAI

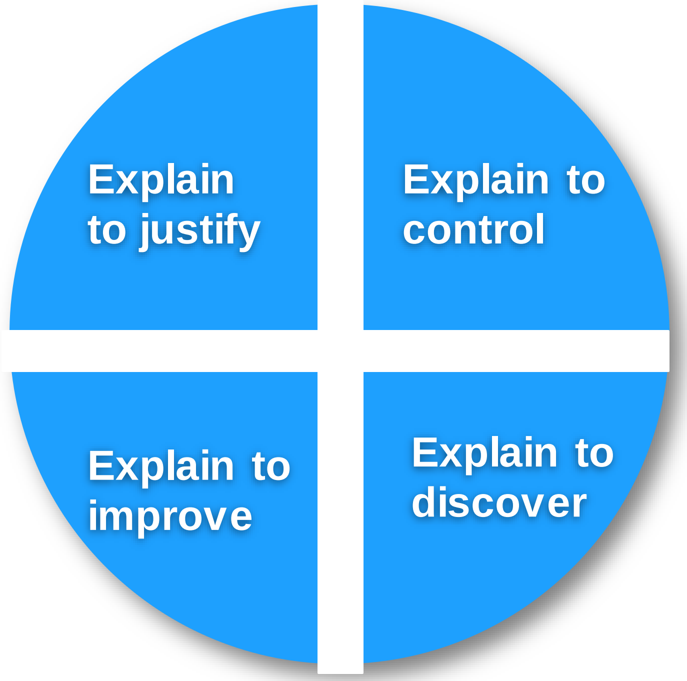
XAI is not a monolithic concept, it reflects several distinct related notions

No.	Related Concept	Description
1	**Interpretable Machine Learning**	ML systems that allow users to understand how inputs are mapped to outputs; focuses on human-understandable logic.
2	**Black-box problem**	When an AI system's internal workings are hidden, making it hard to explain how decisions are made.
3	**Responsible AI**	AI designed with Accountability, Responsibility, and Transparency to align with ethical and social values.
4	**Accurate AI**	Refers to the model's ability to make correct predictions across tasks.
5	**Data science**	Field combining statistics, ML, and data analysis to extract insights and build predictive models.
6	**Social science**	Studies human behavior and society; used in AI to improve explainability and address biases.
7	**Third-wave AI**	AI systems that learn contextually, explain reasoning, and adapt to new tasks (a.k.a. AI 3.0).
8	**Artificial General Intelligence (AGI)**	Highly autonomous AI capable of understanding and performing any intellectual task a human can.

Source: IEEE Xplore (Link) ~ 3,000 citations

Reasons for XAI

The need for explaining AI systems stem from 4 main reasons



No.	Reasons
1	**Explain to justify:** XAI helps provide reasons for decisions, especially unexpected ones, ensuring fairness, trust, and compliance (e.g., GDPR's "right to explanation").
2	**Explain to control:** Understanding system behavior improves visibility, helps detect errors, and enables better debugging and control.
3	**Explain to improve:** Explainable models allow users to understand outcomes and identify areas for improvement, supporting continuous learning and enhancement.
4	**Explain to discover:** Explanations enable knowledge discovery, helping humans learn new insights from AI, including hidden patterns in fields like biology or physics.

Source: IEEE Xplore (Link)

When to apply XAI?

Making AI systems explainable can be costly, requiring significant resources for both development and practical use. Whether explainability is necessary depends on two main factors:

- **Complexity**: Simple AI systems may not need detailed explanations.
- **Risk of Error**: In low-risk applications like advertising, minimal explainability may suffice. In high-risk fields such as medical diagnosis, clear explanations are essential to prevent harm and build trust.

In short, explainable AI is most valuable in areas where the cost of wrong decisions is high.

Domains with potential use of XAI

No.	Domains	Description
1	**Transportation**	In self-driving cars, AI must make quick, life-critical decisions. Errors due to misclassification (e.g., confusing a person for an object) can be fatal. XAI helps explain and prevent such failures.
2	**Healthcare**	AI models in medicine must be interpretable to avoid dangerous outcomes. For example, one model wrongly suggested asthma patients with pneumonia were low risk—XAI would have revealed the flaw.
3	**Legal**	AI used in criminal justice must be transparent to ensure fairness. Cases like *Loomis v. Wisconsin* show the danger of using black-box models in sentencing without explanation.
4	**Finance**	AI in banking and credit scoring must be explainable to comply with regulations and ensure fair decisions. Efforts are underway to generate automated “reason codes” for decisions.
5	**Military**	Military AI systems face ethical challenges similar to healthcare. DARPA’s XAI initiative aims to make autonomous defense systems more transparent and accountable.
6	**Others**	XAI is also useful in cybersecurity, education, entertainment, government, and more—wherever AI decisions impact people and trust is essential.

Technical Challenges of XAI

Accuracy vs. Interpretability

- Advanced models (e.g., DNNs, Random Forests) deliver high accuracy but are difficult to understand.
- Simpler models (e.g., linear regression) are easier to interpret but often less accurate.

Black Box Problem

- Deep models rely on complex, multi-layer structures that obscure decision logic.
- Multiplicity of good models: Similar outcomes may come from different internal paths, making consistent explanations difficult.

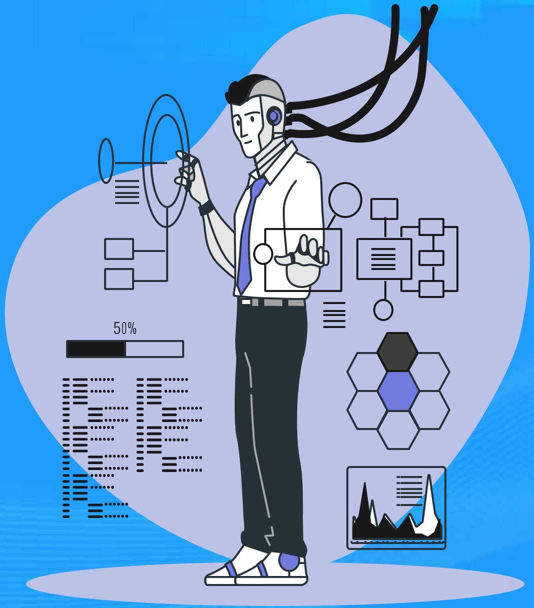
Lack of Standards

- No widely accepted metrics or frameworks to evaluate or compare explainability.

Interdisciplinary Challenge

- Building effective XAI requires expertise from multiple fields: Data Science, AI/ML, Human Sciences, Human-Computer Interaction (HCI)

XAI Approaches



Considerations for XAI

Agnosticity

Indicates whether the XAI method depends on the internal structure of the model (model-specific) or can work with any model (model-agnostic).

Explanation Type

Describes the form of the explanation, like feature importance, rules, examples, visualizations, or natural language descriptions.

Scope

Defines whether the explanation is for a single prediction (local) or the entire model's behavior (global).

Data Type

Refers to the type of input data the explanation method is designed for, such as tabular, text, image, time series, or multimodal data.

Consideration for XAI: by Agnosticity

Agnosticity

Model Agnostic

- LIME
- SHAP
- Partial Dependence Plot (PDPs)

Model Specific

- Attention Mechanism
- Tree interpreter
- CNN Visualizer

Consideration for XAI: by Scope

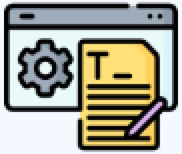
1. Global Interpretability: Understanding the overall behavior and logic of the model across all inputs.

- Helps in **policy-level decisions** (e.g., healthcare, climate change).
- Typically achieved through **simpler models** or **summary techniques**.
- Examples:
  - Additive models for medical prediction.
  - Rule-based models like Bayesian Rule Lists.
  - GIRP – builds interpretation trees from local explanations.
  - Activation maximization to visualize neuron preferences.

2. Local Interpretability: Explaining the reasoning behind **a single prediction** or decision.

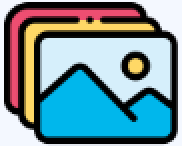
- Supports **case-by-case justifications** (e.g., why was a loan denied?).
- Can be applied to any model, including black-box models.
- Examples:
  - **LIME** – approximates the model locally using simple interpretable models.
  - **Anchors** – generates decision rules for local behavior.
  - **Saliency maps** – visual pixel importance in images.
  - **SHAP** – assigns contributions of each feature to the final prediction.

Consideration for XAI: by Data Type



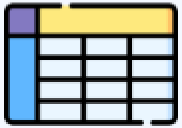
Text

- **Method**: *Sentence Highlighting (SH)*
- **Description**: Highlights the contribution of each word to the prediction.
- **Example Use**: Sentiment analysis, text classification.



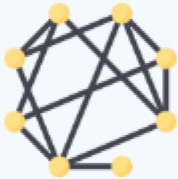
Image

- **Method**: *Saliency Maps (SM)*
- **Description**: Visual heatmaps showing which pixels influence the prediction most.
- **Example Use**: Image classification, object detection.



Tabular

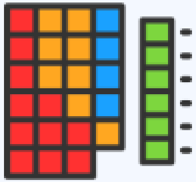
- **Method**: *Feature Importance (FI)*
- **Description**: Assigns a score to each feature indicating its impact on the prediction.
- **Example Use**: Credit scoring, medical diagnosis.



Graph

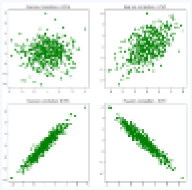
- **Method**: *Node Highlighting*
- **Description**: Scores each graph node based on its influence on the model's prediction.
- **Example Use**: Social network analysis, molecule classification.

Consideration for XAI: by Explanation Type



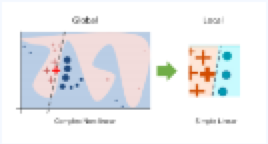
1. Visual Explanations

- *Tool*: Correlation Heatmaps, t-SNE, PCA
- *Goal*: Show variable relationships in the dataset
- *Use case*: Data exploration & understanding model context



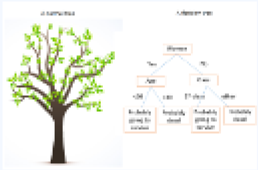
2. Feature Importance

- *Tool*: SHAP, LIME, Permutation Importance
- *Goal*: Quantify how much each feature influences predictions
- *Use case*: Model debugging, trust building



3. Data Point Explanations

- *Tool*: LIME (Local Interpretable Model-agnostic Explanations), Counterfactuals
- *Goal*: Use individual examples to explain prediction decisions
- *Use case*: Understanding specific outcomes (e.g., why was this image classified as a husky?)



4. Surrogate/Simple Models

- *Tool*: Decision Trees, Linear Models
- *Goal*: Approximate a complex model with a simpler one
- *Use case*: Global interpretability of black-box models

XAI Techniques

No.	Technique	Model Agnostic	Data Type	Global | Local	Explanation Type
1	LIME (*)	Yes	Text/Tabular/Image	Local	Surrogate Models
2	SHAP (**)	Yes	Text/Tabular/Image	Local & Global	Feature Importance
3	PDP (***)	Yes	Tabular	Global	Visual
4	CNN Visualizations	No	Image	Local	Visual
5	Counterfactual Explanations	Yes	Text/Tabular/Image	Local	Data Types

(*) LIME - Local Interpretable Model-Agnostic Explanations | **(**) SHAP** - Shapley Additive Explanations | **(***) PDP** - Partial Dependence Plots

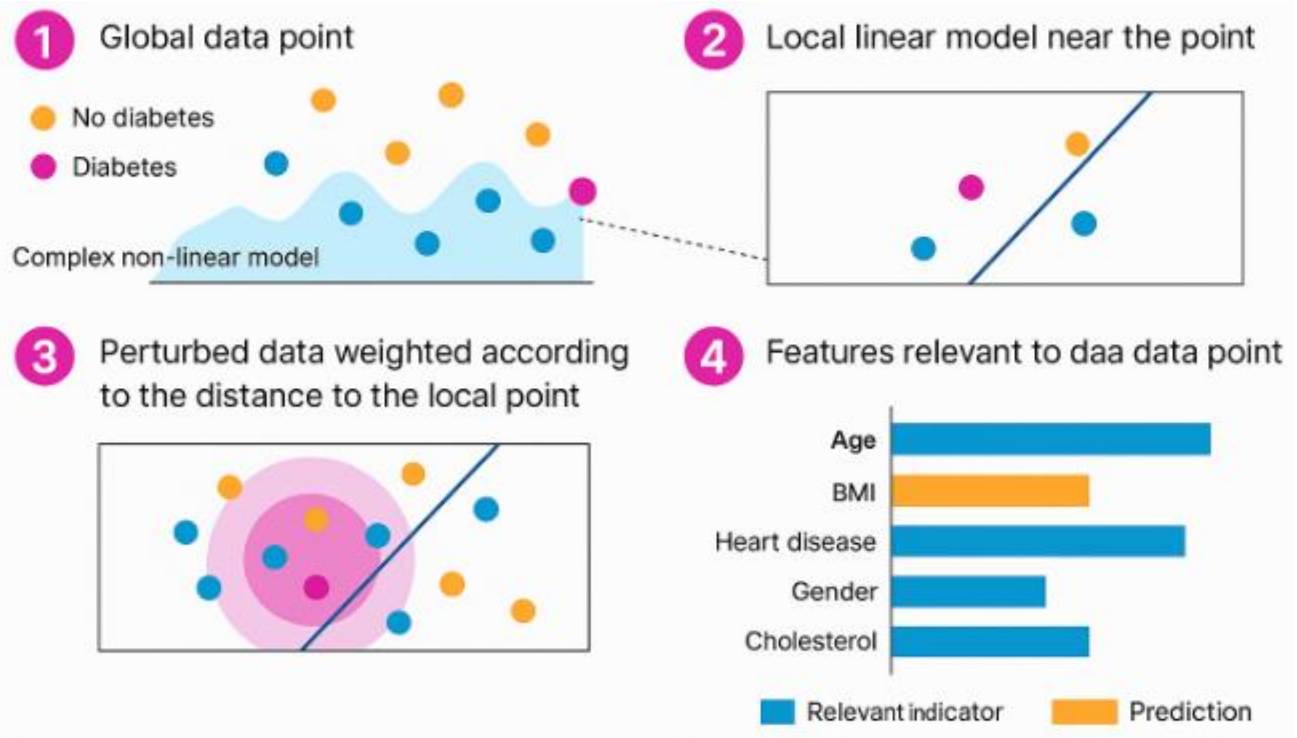
Local Interpretable Model Agnostic Explanations

1. We start with a diabetes classification model. The pink point represents the data sample we want to explain.

2. We zoom in around the pink point to create a local linear model. This local model helps approximate the complex model's behavior near the selected point.

3. We analyze nearby data to see which features have the most influence. Perturbed samples are generated, and their influence is weighted by their closeness to the pink point.

4. We get a ranked list of features based on their impact. This shows which features contributed the most to the model's prediction for the pink point.



Potential Project Application of LIME

Korea	Since 2023

Customer Needs:

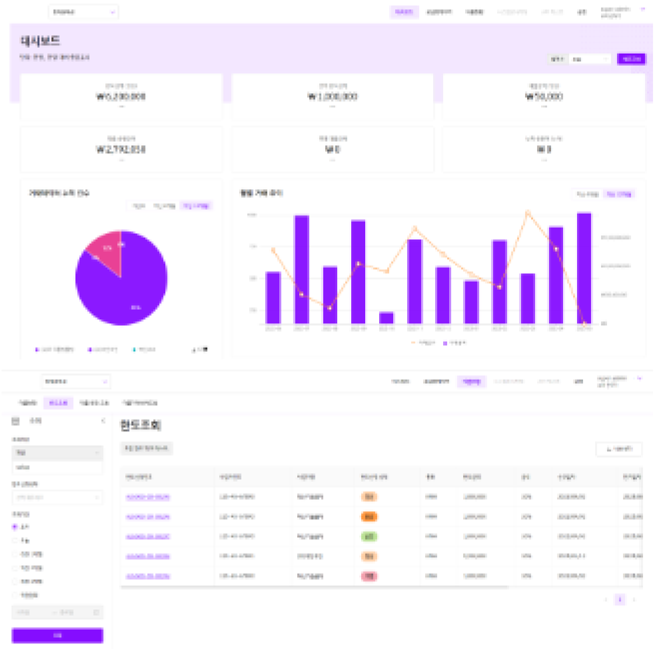
The customer want to build a Credit Scoring method to assess SMEs' debt repayment capability for a P2P Lending Platform of one of the largest bank in Korea.

TMA Values and Achievements:

- Credit Scoring algorithm adjustment improved assessment significantly.
- Our model is implemented into the customer's system pipeline for them to make better decisions.
- Build a model for each financial risk of businesses.
- Data analysis to project transaction revenues from the factory to end-buyers within a Supply Chain.
- Build a scoring system based on the collected insights from the 2 models above.

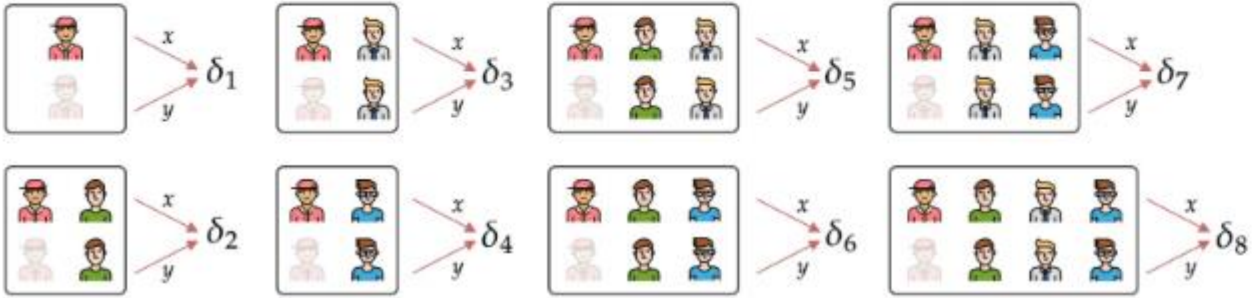
Technologies:

- Cloud Platform: AWS
- Development: Vue.js, Ant Design, Node.js, Nest JS
- Data: Aurora DB, MySQL
- Predictive Analysis: ARIMA, GBM, LSTM, K-means
- Credit evaluation: Linear Regression, NN, SVM
- Blockchain: Hyper ledger Besu



Shapley Additive Explanation

SHAP computes how much each feature contributed to the prediction by considering all possible combinations of features.



The Shapley value for member

is given by:

$$\phi_i = \frac{\delta_1 + \delta_3 + \delta_4 + \delta_5 + \delta_6 + \delta_7 + \delta_8}{8}$$

Partial Dependence Plots

Each subplot (A–D) shows the partial dependence of SHAP values on one feature, helping us understand how changes in a specific feature affect the model's prediction.

(A) Fasting glucose: SHAP values increase after ~116 mg/dL. Higher glucose levels contribute positively to the prediction (e.g., higher risk or likelihood).

=> Implication: The model becomes more confident in predicting a negative outcome (possibly stroke severity or risk) as glucose levels rise above 116 mg/dL.

(B) Initial NIHSS (stroke severity score): SHAP value rises sharply around 6.9 and again near 15.

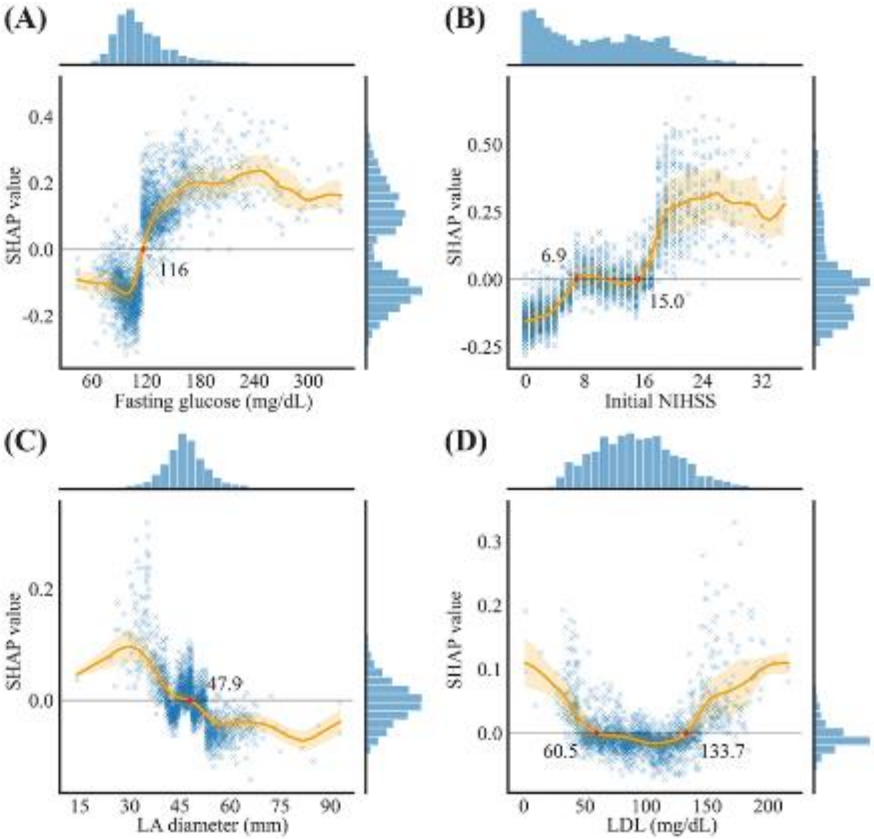
=> Implication: The model assigns a positive weight (risk factor) to higher NIHSS values — it expects worse outcomes with higher stroke severity.

(C) LA diameter (left atrium): SHAP values decrease beyond ~47.9 mm, suggesting a negative contribution (possibly protective or lower risk)

=> Implication: Larger LA diameter might be associated with lower predicted risk in this context.

(D) LDL (bad cholesterol): U-shaped curve. Both very low (< 60.5) and high (> 133.7) LDL values are associated with higher SHAP values (risk)

=> Implication: The model sees moderate LDL as ideal, while extreme values (low or high) increase risk.



Potential Project Application of SHAP & PDP

USA

Since 2019

Customer Needs:

This solution is made for a company providing services and laundry facilities in the US with over 11 million washer and dryer machines. The goal is reducing drying time, predicting relative amount of water remained during cycle, automating and remote control the machine.

TMA Values and Achievements:

- Fuzzy Logic Model: reduce 30% of drying time
- Reinforcement Learning Model: reduce 20%

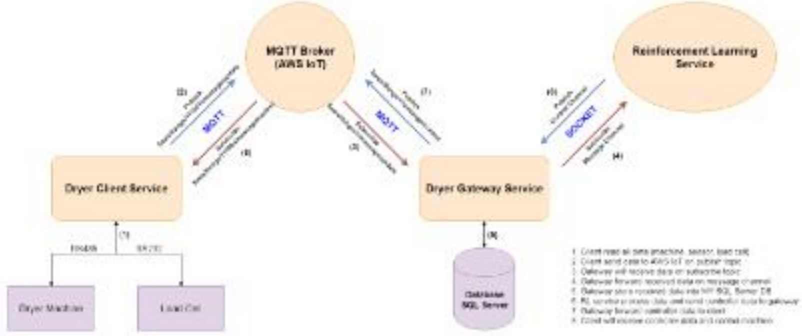
Technologies:

- Machine Learning and AI Models
- Message Queuing Telemetry Transport
- IoT Backend
- Predictive Analytics

Dryer Machine

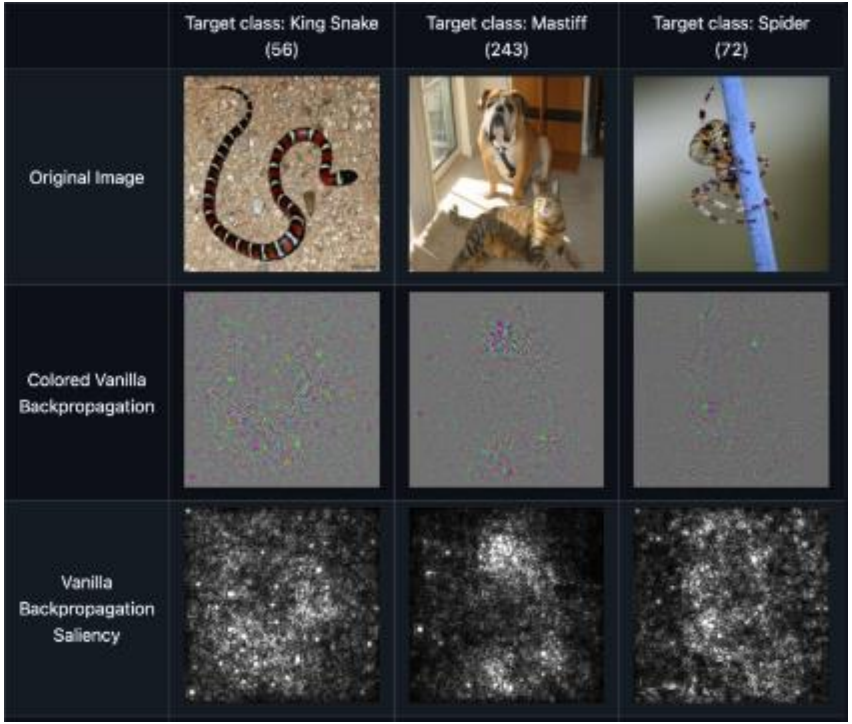
Connect

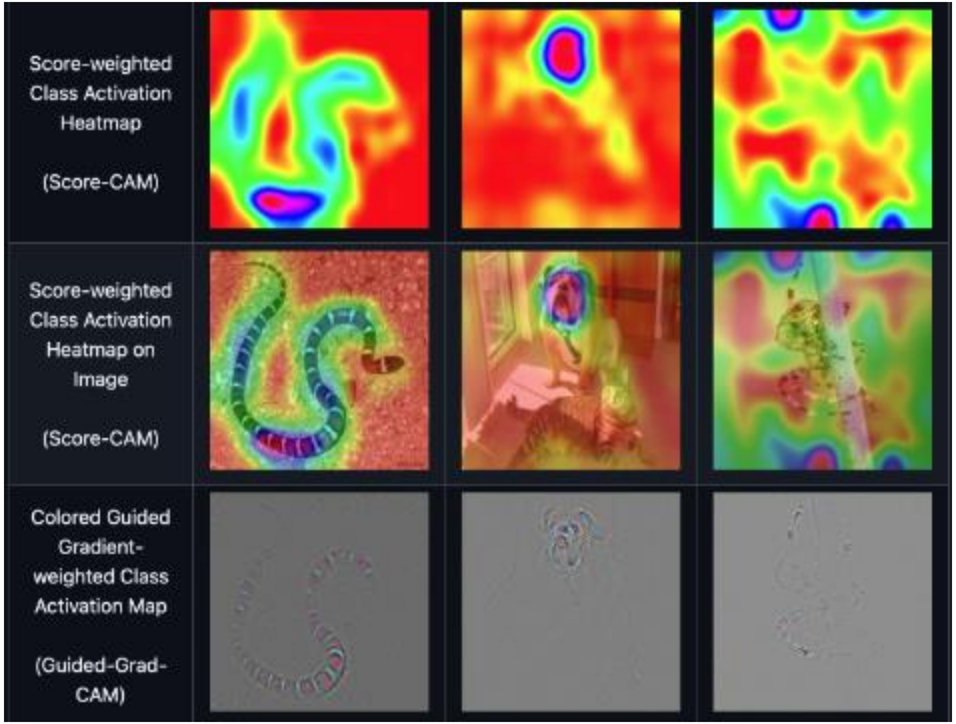
Smart Device Framework



CNN Visualizations

Repository contains a number of convolutional neural network visualization techniques implemented in PyTorch.





<u>Source</u>: Github (Link)

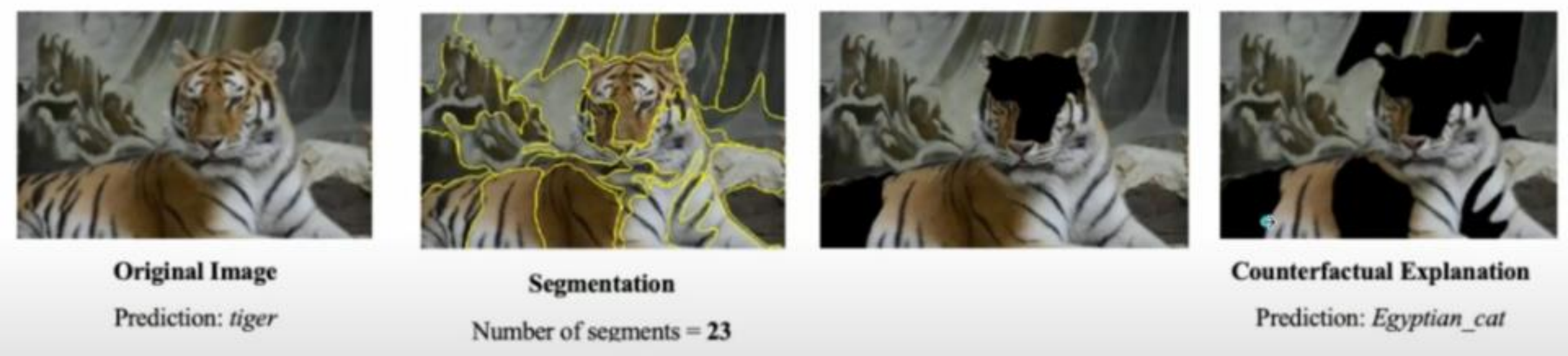
XAI in Lung Cancer

XAI-augmented human knowledge suggests that the lung wall may be indicative of cancer.

Source: NVIDIA on-demand (Link)

Counterfactual Explanation

Make minimal yet measurable changes to see when the model outcome changes



XAI Stories

Other XAI Stories

No.	Story
1	Story House Sale Prices: eXplainable predictions for house sale
2	Story Hotel Booking: eXplainable predictions of booking cancellation and guests coming back
3	Story Hotel Booking Cancellations: eXplainable predictions for booking cancellation
4	Story Uplift Modelling: eXplaining colon cancer survival rate after treatment
5	Story Uplift Modeling: eXplainable predictions for optimized marketing campaigns
6	Story MEPS: Explainable predictions for healthcare expenditures
7	Story MEPS: Healthcare expenditures of individuals
8	Story Lungs: eXplainable predictions for post operational risks
9	Story HELOC Credits
10	Story COMPAS: recidivism reloaded

Source: Link

THANK YOU!